

BATHURST, NB, Canada

WMO: 715740

Lat: 47.631N

Lon: 65.741W

Elev: 59

StdP: 100.62

Time Zone: -4.00 (NAA)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.2	-21.2	-28.2	0.3	-23.1	-25.7	0.4	-20.3	11.5	-7.5	10.3	-7.5	2.1	260	0.595

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	29.4	20.6	27.4	19.7	25.6	18.6	22.0	26.7	21.0	25.6	20.1	23.9	4.6	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	15.3	23.9	19.6	14.4	22.9	18.6	13.5	22.1	64.7	26.6	61.2	25.5	57.9	24.0	26.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.3	7.4	-29.6	32.7	2.7	1.3	-31.6	33.7	-33.2	34.5	-34.7	35.2	-36.6	36.2		
(5)			-29.8	23.9	2.7	0.8	-31.7	24.5	-33.3	25.0	-34.8	25.5	-36.7	26.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-9.8	-9.3	-4.1	2.9	9.8	14.9	19.4	18.3	13.7	7.4	0.8	-5.9		
	DBStd	11.14	6.51	6.08	5.54	3.92	3.93	3.47	3.02	2.82	3.78	3.98	4.92	5.64		
	HDD10.0	2741	613	540	438	215	52	2	0	0	9	100	278	494		
	HDD18.3	5001	872	773	696	463	265	113	23	35	145	338	526	752		
	CDD10.0	892	0	0	1	2	47	151	291	259	119	21	2	0		
	CDD18.3	110	0	0	0	0	2	11	56	36	6	1	0	0		
	CDH23.3	1014	0	0	0	2	41	138	478	272	80	4	0	0		
	CDH26.7	244	0	0	0	0	8	33	127	58	18	1	0	0		
(14) Wind	WSAvg	3.0	3.4	3.5	3.6	3.2	2.9	2.5	2.4	2.3	2.6	3.0	3.1	3.3		
(15) Precipitation	PrecAvg	1086	92	69	85	92	82	81	96	101	73	92	107	111		
	PrecMax	1414	159	118	165	172	160	148	238	219	155	216	202	173		
	PrecMin	857	3	25	26	23	25	40	16	34	26	23	45	44		
	PrecStd	150	42	27	36	45	40	34	49	47	33	52	40	40		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.4	6.6	15.2	21.2	27.4	30.1	32.2	30.8	29.4	22.8	17.5	9.2		
		MCWB	7.9	4.7	10.2	12.6	17.6	19.7	22.1	21.0	21.0	16.4	15.9	7.9		
	2%	DB	4.0	3.2	7.8	15.2	23.7	27.1	29.8	28.3	25.5	19.1	13.5	5.6		
		MCWB	2.8	1.2	4.2	9.1	15.2	18.4	20.9	20.6	18.6	15.0	11.1	4.4		
	5%	DB	2.0	1.2	5.4	12.1	20.3	24.5	27.9	26.2	23.0	16.4	10.3	3.5		
		MCWB	1.0	0.1	2.3	7.1	13.0	16.9	20.4	19.3	17.4	12.8	8.0	2.4		
	10%	DB	-0.1	-0.2	3.4	9.5	17.6	22.2	25.9	24.5	20.9	14.0	7.8	1.5		
		MCWB	-0.9	-1.2	0.9	5.4	11.8	15.9	19.0	18.5	16.5	11.0	6.1	0.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.9	5.1	10.3	14.1	19.1	21.6	23.6	22.7	22.0	18.0	16.3	8.4		
		MCDB	8.5	6.6	13.6	19.5	25.3	27.1	28.5	27.2	25.9	19.8	17.4	9.0		
	2%	WB	3.1	1.8	4.7	9.9	16.5	19.9	22.4	21.6	20.0	15.9	12.0	4.8		
		MCDB	3.7	2.5	6.9	13.6	21.3	24.7	27.1	26.5	23.0	18.6	13.1	5.2		
	5%	WB	0.9	0.2	2.6	7.7	14.5	18.4	21.3	20.6	18.8	13.5	8.2	2.6		
		MCDB	1.9	1.0	4.9	11.5	18.6	22.6	25.7	24.7	22.2	15.7	9.3	3.4		
	10%	WB	-0.9	-1.2	1.1	5.8	12.5	17.0	20.5	19.6	17.1	11.5	6.1	0.8		
		MCDB	-0.2	-0.3	3.1	9.0	16.8	20.7	24.5	23.0	20.3	13.4	7.7	1.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.8	10.2	9.8	9.3	11.4	10.7	11.0	10.7	11.2	9.1	7.6	7.7		
		MCDBR	10.3	10.4	12.1	15.3	16.7	15.8	14.3	13.7	14.6	13.3	11.2	8.6		
	5% WB	MCWBR	9.7	9.0	9.1	9.8	9.5	8.4	7.2	7.2	9.0	9.4	9.4	8.1		
		MCWBR	10.1	9.4	10.8	13.6	14.6	12.7	11.6	11.8	12.6	11.6	10.1	8.3		
(40) Clear-Sky Solar Irradiance	taub		0.277	0.278	0.314	0.379	0.366	0.386	0.418	0.391	0.361	0.331	0.297	0.279		
	taud		2.355	2.325	2.331	2.259	2.387	2.344	2.288	2.358	2.432	2.507	2.532	2.435		
	Ebn at Noon		807	891	907	871	897	878	842	851	847	822	789	766		
	Edh at Noon		70	89	105	126	115	121	126	113	95	76	60	58		
(44) All-Sky Solar Radiation	RadAvg		1.28	2.14	3.35	4.34	5.12	5.53	5.48	4.80	3.71	2.29	1.25	0.93		
	RadStd		0.15	0.24	0.25	0.37	0.47	0.42	0.43	0.30	0.31	0.19	0.16	0.10		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page